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Wettability of Na_2SO_4-NaCl molten salt on CaO-based sorbent surface at high temperature: in-situ experiment and molecular dynamics simulation

Qicheng Chen [a], Qila Sa [a], Nan He [a], Liang Yao [a], Binjian Nie [b] Show more

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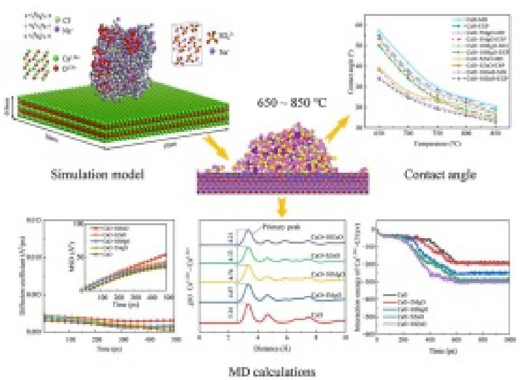
Highlights

- Contact angles of Na_2SO_4-NaCl on the doped-modified CaO surface at high-temperature were measured by in situ experiments.
- Microscopic mechanism of wettability of Na_2SO_4-NaCl on doped-modified surfaces was revealed by MD simulations.
- Doping MgO / ZnO increased substrate pores and improved the wetting of molten salt, enhancing the solid solution formation.

Abstract

Doping alkali metal salts and metal oxides into calcium oxide materials can not only improve the conversion efficiency of calcium looping but also enhance resistance to sintering. In this process, the wettability of molten alkali metal salts on the sorbent surface plays a key role. In this study, the wetting behavior of Na_2SO_4-NaCl binary alkali metal molten salts on CaO doped with MgO and ZnO substrates at high-temperature is investigated by in-situ experiments. The results show that the Na_2SO_4-NaCl salt has good wettability on the various substrates by measuring the contact angle. More importantly, in comparison with the CaO surface, the doping of MgO and ZnO remarkably enhanced the wettability of binary alkali-metal molten salts at high temperatures, and the Na_2SO_4-NaCl molten salt exhibited the best wettability on the ZnO-doped surface. Afterward, the mechanism behind it is revealed deeply by molecular dynamics simulation. The analysis of radial distribution function (RDF), mean square displacement (MSD), and interaction energy indicated that the dopant induced the lattice distortion, resulting in the interaction enhancement between molten salt and substrates, promoting the wettability of molten salt, which is beneficial for the formation of solid solutions.

Graphical abstract



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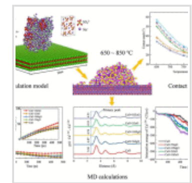
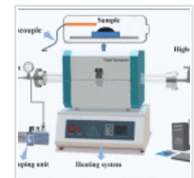
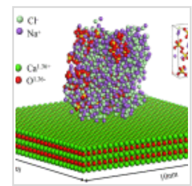
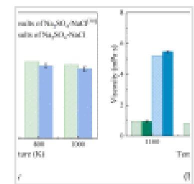
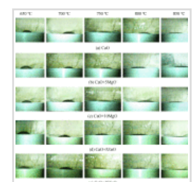
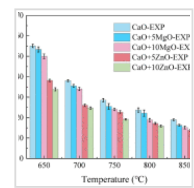

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Tables (1)

Keywords

Binary molten salt; Doped modification; Wettability; In-situ experiments; Molecular dynamics simulation

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1. Introduction

Calcium Looping (CaL) has significant advantages in high-temperature thermochemical energy storage (TCES). It is widely available, inexpensive, and has a high energy storage density, making it an ideal technology for improving the efficiency of thermal energy storage [1,2]. However, as carbonation/calcination increases, sintering occurs in calcium-based materials, reducing the effective conversion rate and thus severely limiting the use of CaL for thermal energy storage [3,4]. Currently, doping modification of calcium-based materials is a significantly promising modification approach to effectively improve the sintering and increase the conversion of the CaL [5].

In recent years, numerous studies have demonstrated that incorporating inert oxides can notably decrease the sintering rate of calcium-based materials. The common inert oxides mainly include MgO [6,7], Al_2O_3 [[8], [9], [10]], ZnO [11,12], and SiO_2 [13], which can significantly enhance the thermal stability of calcium-based compounds by inhibiting the coalescence and crystal growth of calcium oxide particles. The MgO and Al_2O_3, because of their high melting point, prevent agglomeration of calcium oxide grains by physical separation. In addition, ZnO can promote the regeneration of calcium oxide nanoparticles, further improving the sintering resistance [14]. Doped SiO_2 serves as a carrier to mitigate pore clogging because of its unique structure. However, the conversion efficiency can not be entirely increased by the inert oxides since they occupy the reaction sites but do not take part in the carbonation reaction. As a result, the efficiency of CaL is hindered.

Excitingly, recent research demonstrated that doping with metal salt improved the reaction efficiency of CaL, thereby enhancing the performance of thermochemical energy storage. Lee et al. [15] investigated the impact of Na_2CO_3 doping on the kinetics of CO_2 adsorption by calcium oxide sorbent. The synthesized sorbent developed a $Na_2Ca(CO_3)_2$ double salt, which altered the carbonation reaction pathway of the sorbent and enhanced its diffusion kinetics. The formation of the double salt made the sorbent easier to regenerate, thus improving its cyclic stability. Kong et al. [16] investigated that doping Na_2SO_4 promoted the effective conversion rate of CaL primarily by improving the kinetics of reactions. Xu et al. [17] mixed CaO and KCl using the wet dispersion method. The results showed that the CO_2 adsorption capacity of KCl-doped CaO significantly increased and was twice as high as that of untreated CaO. Huang et al. [18] revealed that the molten Na_2CO_3 and Na_2SO_4 promoted the formation of carbonate ions (CO_3^{2-}) due to the high concentration of oxygen ions (O^{2-}) in molten alkali metal salt. This led to the rapid formation of $CaCO_3$ and the facile regeneration of particles at high temperatures. Thus, the adsorption rate of CO_2 could be improved. It should be noted that alkali metal salt did not function as an anti-sintering agent. On the contrary, as the number of carbonation and calcination rose, they might hasten the sintering process of the sorbent [16].

In our previous study, a novel calcium matrix composite was prepared using the Na_2SO_4-NaCl-ZnO co-doping method. The results showed that the Na_2SO_4-NaCl-ZnO composite had a higher conversion rate and energy storage density than CaO. The composite material achieved a fast reaction rate and good cyclic stability [19]. In the process, it is remarkable that the effect of the alkali metal salt in CaL relies on their good wettability on the surface of the calcium-based sorbent. However, the wetting behavior of molten alkali metal salt in CaL has not yet been fully understood, and the microscopic mechanisms behind it are even less known.

This work aims to probe the wettability of Na_2SO_4-NaCl binary alkali metal molten salt on CaO doped with MgO and ZnO substrates at high temperatures. In-situ experiments investigated the contact angle of Na_2SO_4-NaCl molten salt on various substrates. The effects of temperature, dopant type, and dopant concentration on wettability were discussed in detail. Furthermore, molecular dynamics simulation revealed the micro-mechanisms behind it. The results of this research offer a theoretical basis and practical direction for the development and use of sorbent materials in upcoming CaL applications.

2. Experimental procedure and simulation details

2.1. Materials and sample preparation

CaO (Tianjin Guangfu Technology Development Co., Ltd., China, 99.99%), Na_2SO_4 (Xilong Scientific Co. Ltd., China, 99.99%), NaCl (Xilong Scientific Co. Ltd., China, 99.99%), MgO (Xilong Scientific Co. Ltd., China, 99.99%) and ZnO (Xilong Scientific Co. Ltd., China, 99.99%) are analytical purity and can be utilized without additional treatment.

The preparation of Na_2SO_4-NaCl binary alkali metal salt samples involved several key steps. Initially, Na_2SO_4 and NaCl powders were blended entirely with deionized water at a molar ratio of 0.464:0.536 to obtain the mixture of binary alkali metal salt. The mix of binary alkali metal salt was then continuously stirred for 2h using a magnetic stirrer to ensure proper blending. Following this, the binary alkali metal salt was placed in a tube furnace, where it was dried for 2h at 500 °C under an N_2 atmosphere. Lastly, after adequate grinding, the final Na_2SO_4-NaCl binary alkali metal salt sample was produced, weighing approximately 1 g. When the molar ratio of Na_2SO_4: NaCl is 0.464: 0.536, the Na_2SO_4-NaCl has the lowest melting point of 620 °C [19].

In order to obtain an effective conversion rate, this method was used to obtain a composite material with a combination of Na_2SO_4+NaCl-CaO referred to as NN-CaO, which was prepared with a molar ratio of 0.464:0.536:5. Subsequently, 0.25 mol and 0.5 mol of MgO and ZnO were doped into NN-CaO. The resulting co-doped composite sorbents were named NN-CaO+5MgO, NN-CaO+10MgO, NN-CaO+5ZnO, and NN-CaO+10ZnO.

To prepare doped modification oxide substrates, ZnO and MgO powders were thoroughly mixed with CaO powder in molar ratios of 10:0.5 and 10:1, respectively. The uniformly doped compositions, referred to as CaO+5MgO, CaO+10MgO, CaO+5ZnO, and CaO+10ZnO, were created through a grinding process followed by high-pressure extrusion at 45 MPa. This procedure produced cylindrical substrates that measured 4 cm in diameter, 1 cm in thickness and weighed approximately 20 g [20].

2.2. Experimental procedure

In this study, we designed a high-temperature in-situ experiment employing the seated drop method to explore the wetting behavior of the Na_2SO_4-NaCl binary alkali metal molten salt on calcium oxide substrates, as shown in Fig. 1. The experimental system included core components such as a high-speed camera, a heating system, and an evacuation device. The experimental system ensured that the samples were heated and processed under an inert gas environment through a gas purification system to avoid oxidation or decomposition phenomena. The sample bench was equipped with a B-type thermocouple as the temperature sensing element and a tube furnace as the heating system. Throughout the experiments, the sample-substrate set was placed in a horizontal position on the support platform and then gradually heated to the predetermined temperatures (650 °C, 700 °C, 750 °C, 800 °C and 850 °C) at a rate of 20 °C/min. Once the samples attained the specified temperatures, the N_2 supply was turned off, and a high-speed digital camera was turned on to capture the droplet features.

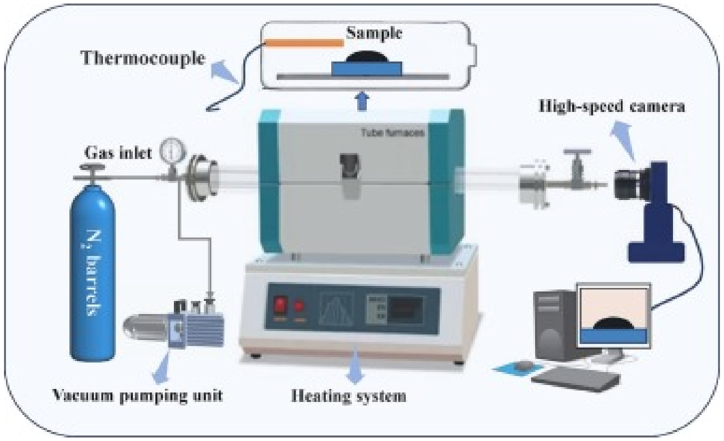

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Fig. 1. Schematic diagram of the experimental setup.

To guarantee the reliability and precision of data, each experiment was replicated at least two times to examine its repeatability. When the discrepancy between the two results fell within the ±3° range, the average value was adopted as the final measurement.

The thermogravimetry (TG) method was employed to analyze the carbonation and calcination of calcium-based composite sorbents. Initially, a sample of 25 mg of the composite was placed in a crucible and subjected to heating from ambient temperature to 700 °C at a rate of 20 °C/min in an N_2 atmosphere flowing at 50 ml/min. Following this, the atmosphere was changed to CO_2 at the same flow rate, and the temperature was elevated from 700 °C to 850 °C at a rate of 5 °C/min to ensure thorough reaction with the material. Once the temperature reached 850 °C, the environment was switched back to N_2 at a flow rate of 50 ml/min and maintained for 10 min to achieve complete calcination of the sample. To evaluate the cycling performance of the calcium-based composite sorbents, the effective conversion rate X_{ef} is computed by Eq. (1), and the energy storage density (E)

is computed by Eq. (2).

$$X_{ef} = \frac{\Delta m}{m_0} \cdot \frac{W_{\mathrm{CaO}}}{W_{\mathrm{CO_2}}} \tag{1}$$

$$E = X_{ef} \cdot \frac{\Delta H}{W_{\mathrm{CaO}}} \tag{2}$$

where Δm represents the mass increase during carbonization, m_0 represents the mass of the CaO in the composite sorbents before the carbonization reaction, W_{CaO} represents the molar mass of CaO (56 g/mol), and $W_{\mathrm{CO_2}}$ represents the molar mass of CO_2 (44 g/mol), ΔH=177.8 KJ/mol represent the standard enthalpy of the carbonation reaction, E represents the energy storage density (KJ/Kg).

2.3. Simulation details

2.3.1. Simulation model

Fig. 2 shows the molecular models utilized in this research, which include the relative positions and interactions of the Na_2SO_4-NaCl binary solid salt and CaO substrates to reveal the mechanism behind the wettability of Na_2SO_4-NaCl molten salt on the surface of CaO-based sorbent. Firstly, the crystal cells of Na_2SO_4 and NaCl were established. The lattice parameters of Na_2SO_4 crystal were a=0.5863 nm, b=1.2304 nm, and c=0.9821 nm, with all the angles of α=β=γ=90°. The length of the S-O bond was 0.15 nm, and the angle of the O-S-O bond was 109.5°. The NaCl crystals were hypercubic with lattice parameters of a=b=c=0.562 nm and all angles of α=β=γ=90°. Then Na_2SO_4-NaCl binary solid salt crystals with the volume of 3.377×3.377×3.377 nm^3 were obtained by mixing and annealing equilibrium with a molar ratio of 0.464:0.536. The number of Na_2SO_4-NaCl binary mixed salt atoms was 991. Subsequently, a CaO substrate was constructed. The CaO crystals also showed a hypercubic structure, with lattice parameters of a=b=c=0.48105 nm and angles of α=β=γ=90°, and the CaO atoms were 9244. The CaO(100) has been selected as the exposed crystal plane due to its relatively low surface energy and excellent high temperature stability [21,22]. Subsequently, a 10×10×10 nm^3

vacuum layer was created. Meanwhile, to systematically investigate the effect of MgO and ZnO metal oxides doped in CaO on the wetting behavior, the X(Mg, Zn) atoms in them were substituted for the Ca atoms at 5% and 10% doping rates to obtain the substrate models named CaO+5MgO, CaO+10MgO, CaO+5ZnO, and CaO+10ZnO. In addition, we chose the model of the MgO/ZnO exposed to the crystal plane because the MgO/ZnO was exposed on the surface of the substrate in our experiments.

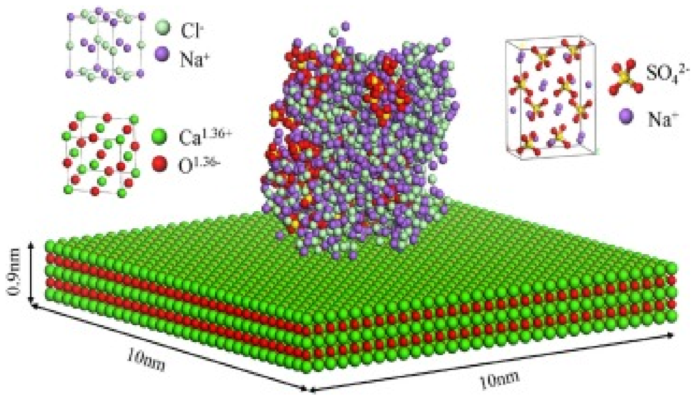

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Fig. 2. Modeling of Na_2SO_4-NaCl binary salt on CaO substrate.

2.3.2. Potential function

In this study, the Lennard-Jones (L-J) potential energy function calculates and simulates molecular and inter-atomic interactions within the system [23]. The expression of the potential energy function is shown in Eq. (3):

$$U_{ij} = \frac{1}{4\pi\varepsilon_0}\frac{q_i q_j}{r_{ij}} + 4\varepsilon_{ij}\left(\frac{\sigma_{ij}^{12}}{r_{ij}^{12}} - \frac{\sigma_{ij}^{6}}{r_{ij}^{6}}\right) \tag{3}$$

where the subscripts *i* and *j* represent the *i*th and *j*th particles, *q* represents the particle charge, r_{ij}、 ε_{ij} and σ_{ij} represent the distances between the *i*th and *j*th particles, respectively, and ε_0 represents the vacuum permittivity. The interactions between different particles are calculated using the mixing law, where the following parameters are obtained from the Lorentz-Berthelot mixing law [23]. Zhao et al. [24,25] used the L-J potential along with long-range Coulomb interactions to calculate MgO, ZnO, and CaO, proving that the potential functions are accurate. Therefore, no further validation is performed in this work. The specific parameters of the potential function are given in Table 1[26]:

$$\varepsilon_{ij} = \sqrt{\varepsilon_i \varepsilon_j} \tag{4}$$

$$\sigma_{ij} = \frac{1}{2}(\sigma_i + \sigma_j) \tag{5}$$

Table 1. Parameters of LJ potential function.

Molecule	**Atom**	***ε* (KCal/mol)**	***σ* (nm)**	**charge (*e*)**
Na_2SO_4	Na	0.096531	0.2580	1.00
	S	0.241213	0.3550	2.00
	O	0.023060	0.3916	−1.00
NaCl	Na	0.130100	0.2350	1.00
	Cl	0.100100	0.4400	−1.00
CaO	Ca	5.0298×10^{-6}	0.5909	1.36
	O	1.554000	0.35532	−1.36
MgO	Mg	9.0298×10^{-7}	0.62484	1.36
	O	1.554000	0.35532	−1.36
ZnO	Zn	0.001500	0.5822	1.36
	O	1.554000	0.35532	−1.36

2.3.3. Simulation methods

The simulation utilized the open-source molecular dynamics software package LAMMPS. Regarding the studied working condition, the whole simulation process could be split into two stages: the equilibrium stage and the dynamics stage. In the equilibrium stage, the optimization of the model ensured that all atoms were in equilibrium. The simulation domain was set to $10\times10\times10\,nm^3$, and the CaO-based substrate was fixed at its initial position and treated with periodic boundary conditions. The Velocity-Verlet algorithm [27] was used to determine the position, velocity, and acceleration of each atom, and the solution of the Newtonian equations of motion simulated the kinetic behavior of the system. A time step of 1 fs was set to guarantee calculation accuracy, the truncation radius was set to 10 Å, and the neighborhood was set to 2 Å to improve the calculation efficiency. The PPPM [28] method calculated the Coulomb interaction in the potential function. The canonical ensemble (NVT) system was always used to calculate during the simulation. The temperature was set at 300K. A 200 ps simulation run was carried out to acquire the equilibrium model of the salt on the substrate surface. Then, the equilibrium melt model with temperature targets of 650 °C, 700 °C, 750 °C, 800 °C and 850 °C was obtained by the heating process. In the dynamics stage, the total simulation time was set to 800 ps, and the relevant calculation data was output every 1000 steps (every 1 ps). In this way, the system evolution could be followed in real time, and important dynamic information could be obtained.

2.3.4. Verification

To confirm the precision of the potential functions, the density of the Na_2SO_4-NaCl binary salt and the viscosities of NaCl and Na_2SO_4 were computed. The density refers to the formula ρ=m/V. The viscosity η was calculated by nonequilibrium molecular dynamics (NEMD), which is shown in Eq. (6) [29]:

$$\eta = -\frac{\sum_{transfer} m(V_{xhot} - V_{xcold})}{2tL_xL_y(\partial V_X/\partial Z)} \tag{6}$$

where the numerator term represents the momentum exchange quantity of the simulated system within time t, t is the momentum exchange time, and L_x and L_y represent the length of the simulated box in the x and z directions, respectively. $\partial Vx/\partial Z$ represents the velocity gradient of the x-direction velocity of the z-axis direction. The density and viscosity results are presented in Fig. 3. They demonstrate that MD simulations were in good agreement with Refs. [[30], [31], [32]], indicating the accuracy of the potential function.

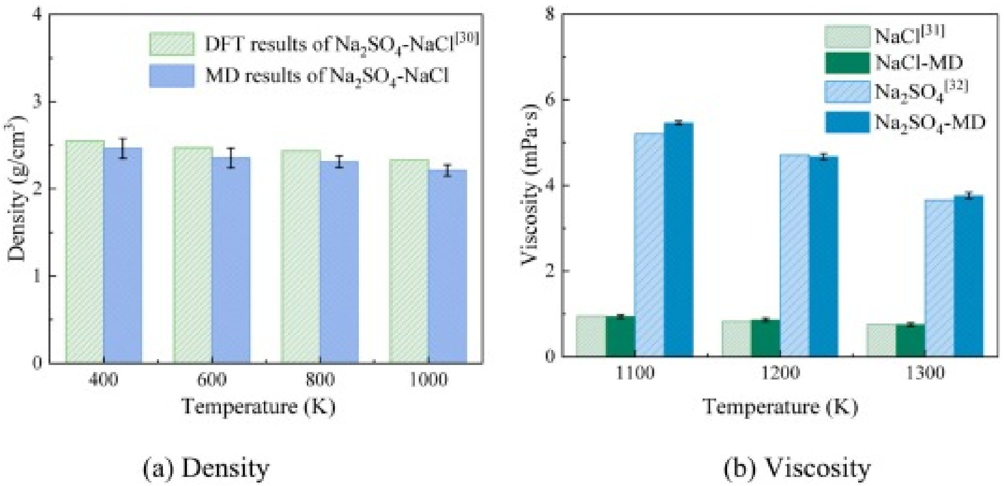

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Fig. 3. Densities and viscosities of alkali metal salt (a) Density (b) Viscosity.

3. Results and discussion

3.1. Wettability of Na_2SO_4-NaCl on the surface of the CaO-based substrate

A high-speed digital camera constantly recorded the wetting behavior of the Na_2SO_4-NaCl molten salt on the surface of the substrate during the entire experiment. The recording commenced when the contact angle between the Na_2SO_4-NaCl molten salt and the substrate showed no significant variation. Additionally, we chose the state after 60s of heat preservation as the endpoint of the experiment. At this endpoint, we computed the averaged contact angle, which allowed us to confirm the reliability and stability of our experimental findings. Fig. 4 shows the wetting behavior of the Na_2SO_4-NaCl molten salt on the surface of CaO, CaO+5MgO, CaO+10MgO, CaO+5ZnO, and CaO+10ZnO.

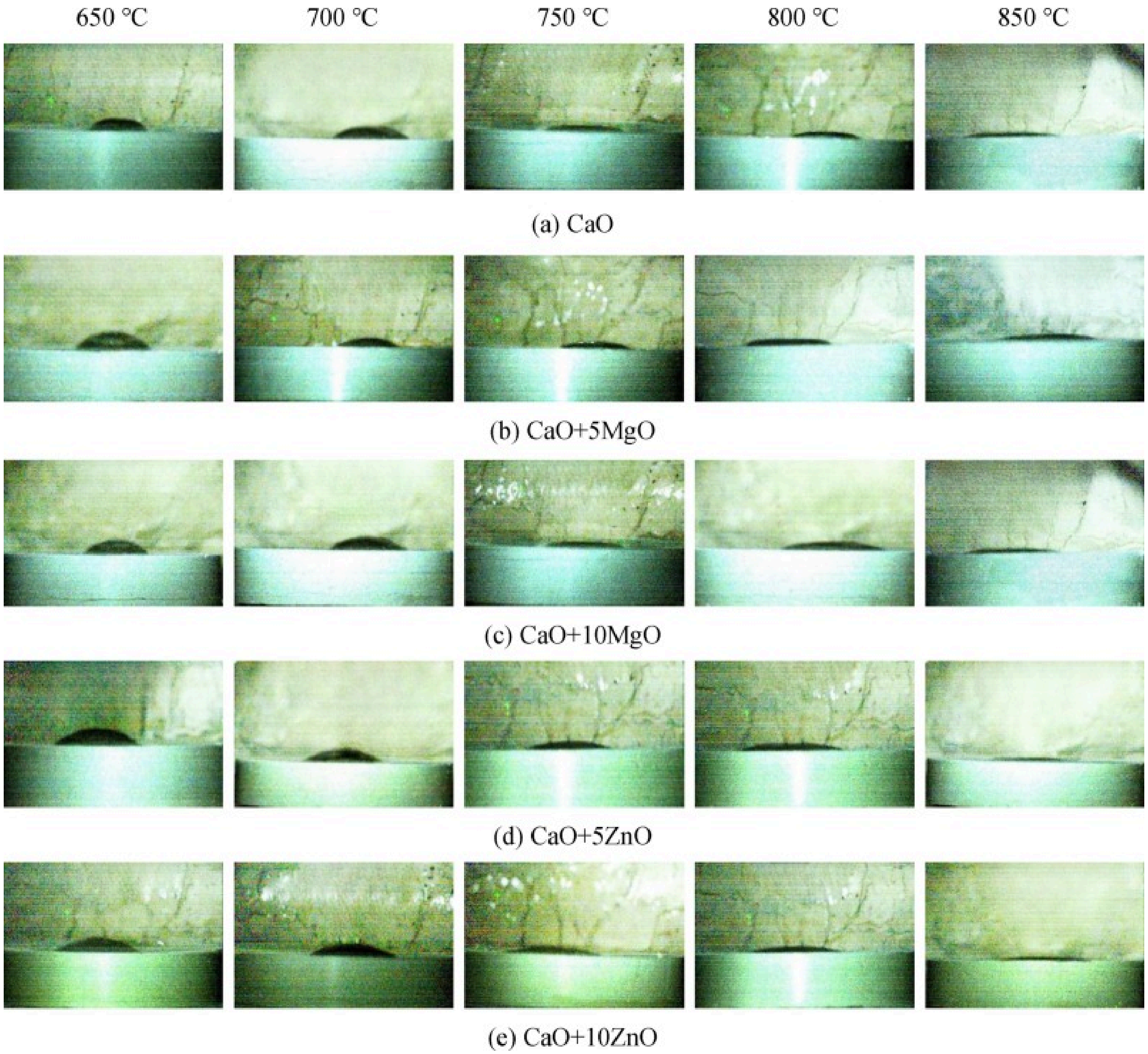

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Fig. 4. Snapshot of the Na_2SO_4-NaCl wetting on the surfaces of (a) CaO, (b) CaO+5MgO, (c) CaO+10MgO, (d) CaO+5ZnO and (e) CaO+10ZnO at 650–850°C.

The contact angles of Na_2SO_4-NaCl molten salt on CaO, MgO-doped, and ZnO-doped substrates were measured in this work according to Ref. [33]. As depicted in Fig. 5, the contact angles on all surfaces steadily declined with the temperature increases. At the 650–850°C temperature range, the contact angle on the CaO surface decreased from 55° (650°C) to 19° (850°C), the contact angle on the CaO+5MgO surface decreased from 53° (650°C) to 16° (650°C), the contact angle on the CaO+10MgO surface decreased from 50° (650°C) to 15° (850°C), the contact angle on the CaO+5ZnO surface decreased from 38° (650°C) to 14° (850°C), and the contact angle on the CaO+10MgO surface decreased from 34° (650°C) to 13° (850°C), which clearly show that with the temperature increased the wettability of the Na_2SO_4-NaCl molten salt was enhanced.

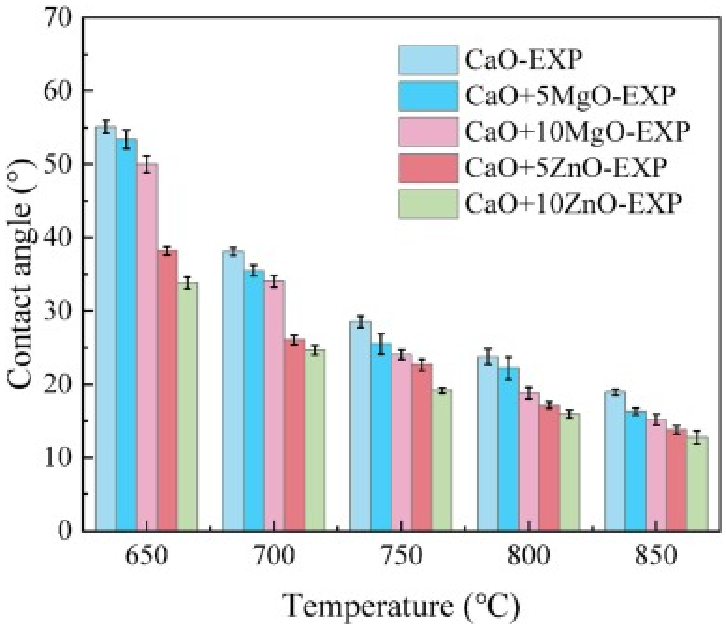

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Fig. 5. Contact angles of Na_2SO_4-NaCl molten salt on CaO-based substrate.

In addition, at the same temperature and same doping concentration, compared to CaO, the contact angles on the surface of MgO-doped and ZnO-doped substrates were reduced to different degrees, with ZnO dopants significantly better than MgO dopants in reducing the contact angle. This indicates that doping MgO and ZnO into the CaO substrate could effectively enhance the wettability of the molten salt. In particular, doping ZnO into the CaO substrate was most effective in increasing the wettability of molten salt. Furthermore, it was found that on the MgO-doped substrate, the contact angles decreased from 53° (650°C) to 50° (650°C) and 16° (850°C) to 15° (850°C) as the MgO doping concentration increased from 5% to 10%. Similarly, on the ZnO-doped substrate, as the ZnO doping concentration rose from 5% to 10%, the contact angles decreased from 38° (650°C) to 34° (650°C) and from 14° (850°C) to 13° (850°C). This trend indicates that the wettability of the Na_2SO_4-NaCl molten salt on its substrate was enhanced with increased doping content for MgO and ZnO in the CaO substrate.

3.2. X_{ef} and energy storage density (E) of the calcium-based composite sorbents

Fig. 6 shows the X_{ef} and energy storage density (E) averaged over the ninth and tenth cycles on the CaO, MgO-doped, and ZnO-doped composite sorbents. For the NN-CaO, the X_{ef} is 47.43%, and the E is 1271.4 KJ/Kg. For the NN-CaO+5MgO, the X_{ef} is 48.53%, and the E is 1262.5 KJ/Kg. For the NN-CaO+10MgO, the X_{ef} of is 51.45%, and the E is 1300.2 KJ/Kg. For the NN-CaO+5ZnO, the X_{ef} is 52.66%, and the E is 1329.8 KJ/Kg. For the NN-CaO+10ZnO, the X_{ef} is 56.74%, and the E is 1354.5 KJ/Kg. It can be seen that the X_{ef} and the E of MgO-doped and ZnO-doped composite sorbents increase with the rise of the doping ratio, and it can also be found that with the same doping ratio, the X_{ef} and the E of ZnO-doped composite sorbents are higher than that of MgO-doped composite sorbents.

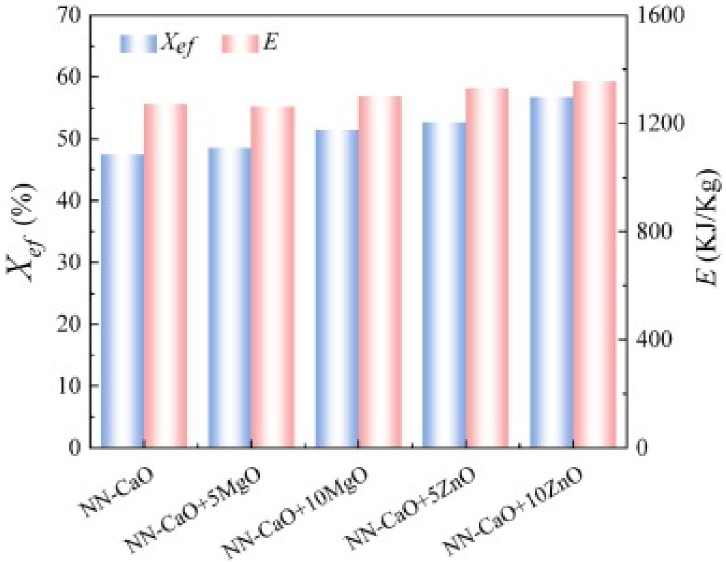

Fig. 6. X_{ef} and E of the averaged over the ninth and tenth cycles of composite sorbents.

Fig. 7 shows the quantitative relationship between improved wettability and higher X_{ef} and E. The relationship between the wettability and the X_{ef} follows the equation (X_{ef}=0.35 θ^2-12.72 θ+162.76, R^2=0.96). The relationship between the wettability and the E follows the equation (E = 4.65 θ^2-163.96 θ+2706.5, R^2=0.91). These results indicate that the enhancement of the X_{ef} and E is similar to the trend of the wettability of Na_2SO_4-NaCl molten salt, which further proves that the excellent wettability of Na_2SO_4-NaCl binary alkali metal molten salt plays a key role in the effective conversion rate of CaL.

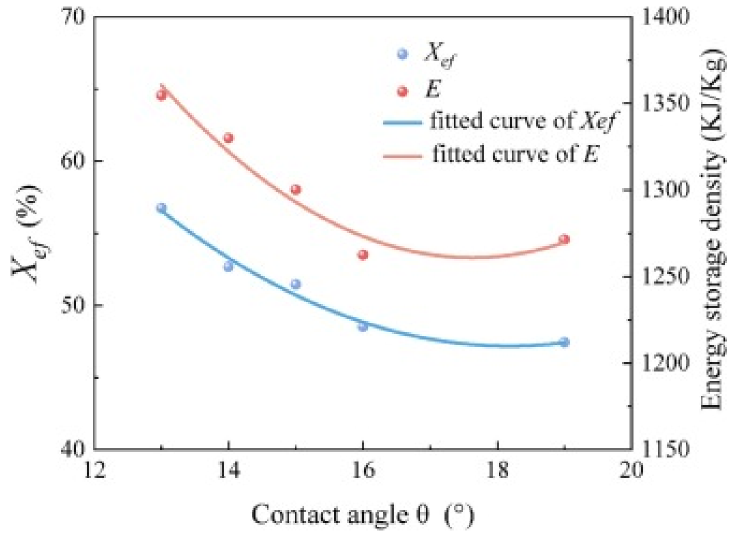

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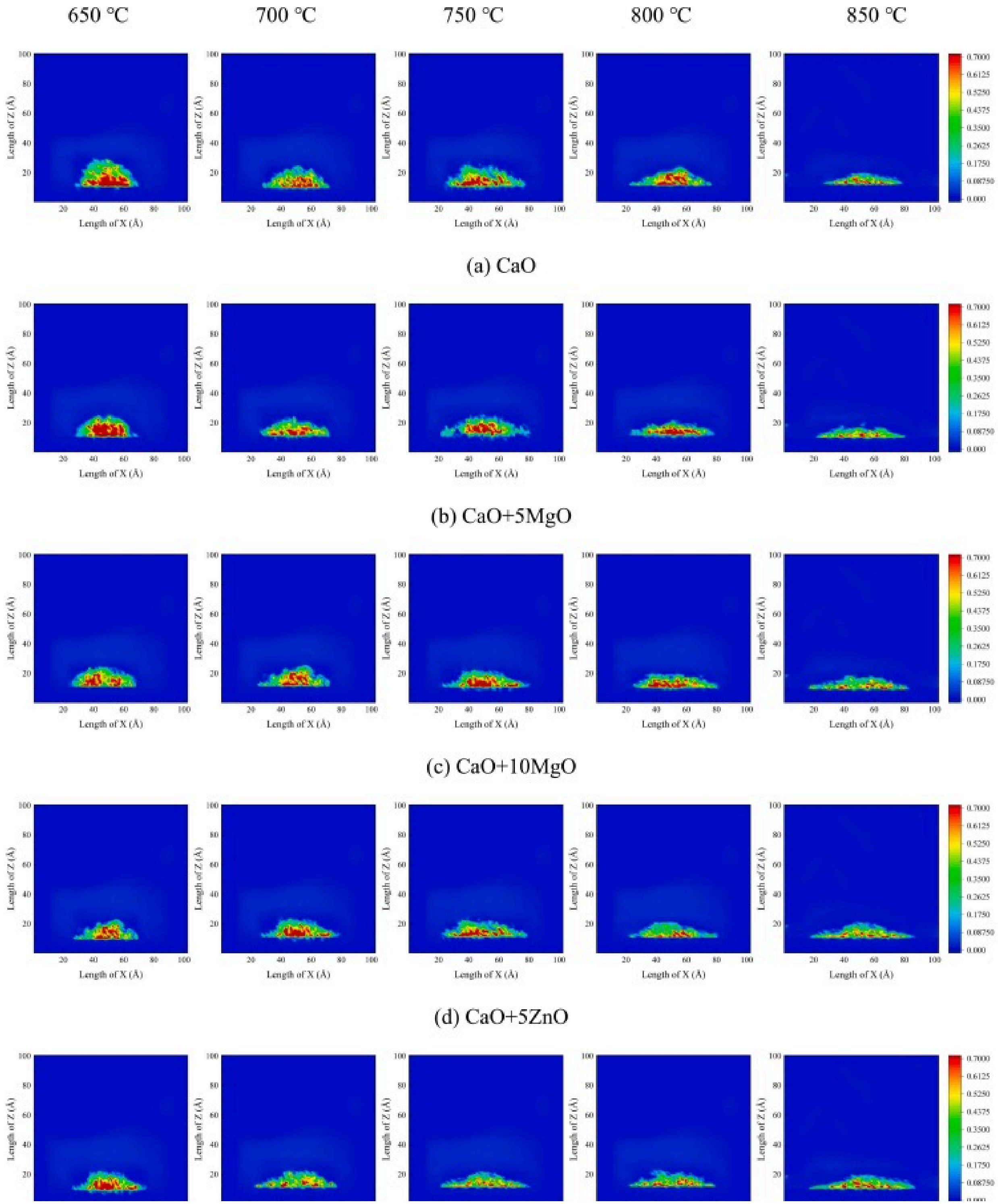
Fig. 7. The quantitative relationship between wettability with X_{ef} and E.

3.3. Micro-mechanism of the wettability

Fig. 8 illustrates the density distribution of Na_2SO_4-NaCl molten salt on the surfaces of CaO, MgO-doped, and ZnO-doped substrates. Comparative analysis showed that the MD simulation results fully agreed with the experimental results, which were further verified by molecular dynamics simulations.


650 ℃
700 ℃
750 ℃
800 ℃
850 ℃
Length of Z (Å)
Length of X (Å)
0.7000
0.6125
0.5250
0.4375
0.3500
0.2625
0.1750
0.08750
0.000
(a) CaO
(b) CaO+5MgO
(c) CaO+10MgO
(d) CaO+5ZnO

(e) CaO+10ZnO

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Fig. 8. Density distribution of the Na_2SO_4-NaCl on the surfaces of (a) CaO, (b) CaO+5MgO, (c) CaO+10MgO, (d) CaO+5ZnO and (e) CaO+10ZnO at 650–850°C.

To further prove the above conclusion, we compared the experimentally measured contact angle with the simulated contact angle, as shown in Fig. 9. The maximum error between the experimentally measured and simulated contact angles did not exceed 10%. It also showed the rationality of the potential function (calculation method) chosen in this work.

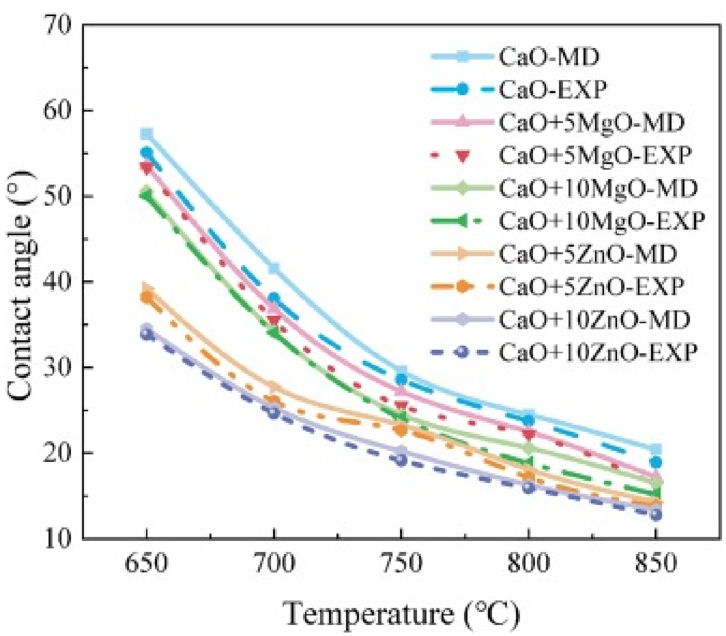

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Fig. 9. Contact angle of Na_2SO_4-NaCl on the surface of the CaO-based substrates.

3.3.1. Mean square displacement (MSD) and self-diffusion coefficient

Fig. 10 shows the mean square displacement (MSD) and self-diffusion coefficient of the Na_2SO_4-NaCl molten salt as it wets the surface of the CaO, MgO-doped, and ZnO-doped substrates at different temperatures. The MSD is computed using Eq. (7) [34].

$$MSD = \left\langle \left| r_i(\tau) - r_i(0) \right|^2 \right\rangle \tag{7}$$

where $r_i(0)$ and $r_i(\tau)$ represent the position of the ith particle at moments 0 and τ, and < > represents averaging over all particles. The self-diffusion coefficient of Na_2SO_4-NaCl molten salt can be derived from the MSD. It is 1/6 of the slope of the MSD curve, calculated by Eq. (8).

$$D_{self} = \frac{1}{6} \lim_{\tau=0} \frac{d}{d\tau} \left\langle \left| r_i(\tau) - r_i(0) \right|^2 \right\rangle \tag{8}$$

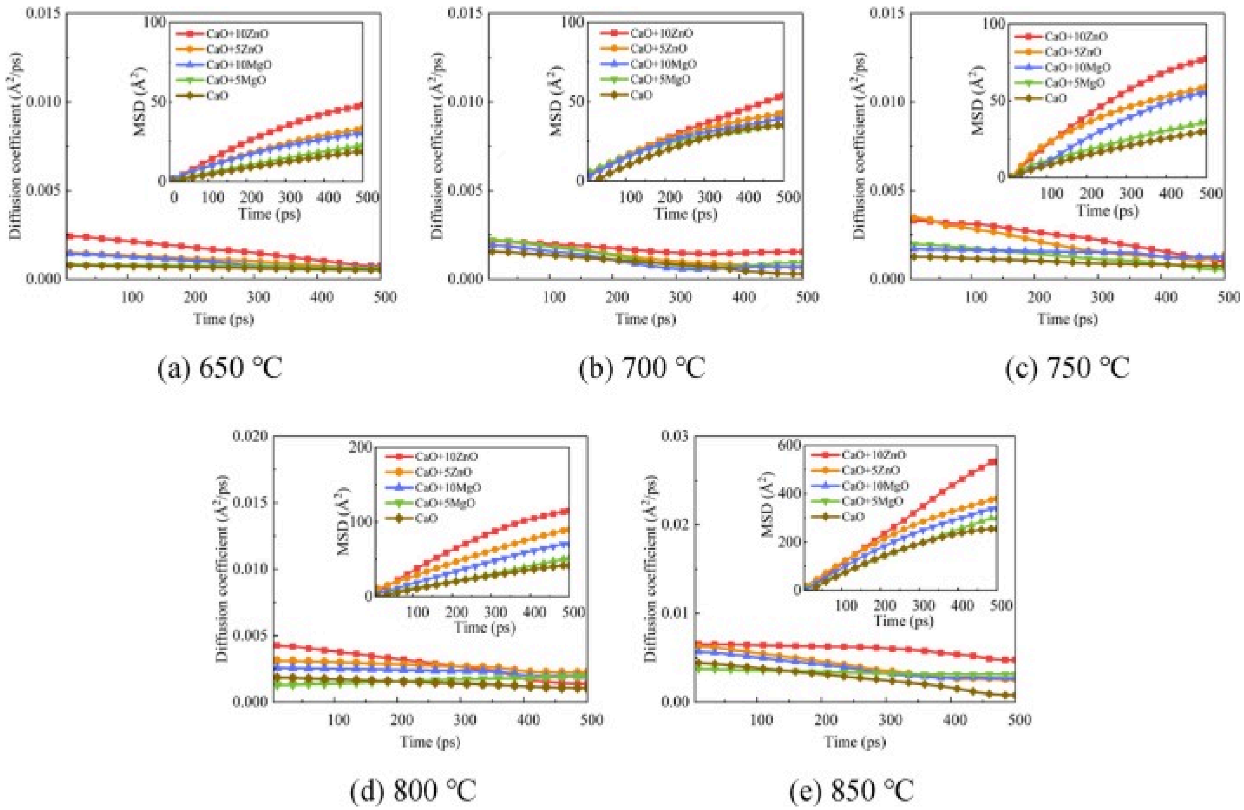

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Fig. 10. MSD and self-diffusion coefficient of molten Na_2SO_4-NaCl at (a) 650 ℃, (b) 700 ℃, (c) 750 ℃, (d) 800 ℃ and (e) 850 ℃.

As shown in Fig. 10, the MSD and diffusion coefficient of the Na_2SO_4-NaCl molten salt increase with rising temperatures, which indicates that the molten salt diffusion properties increase progressively. Further comparison of MSD and diffusion coefficient of Na_2SO_4-NaCl molten salt shows that the diffusivity of molten salt was better than that of the CaO substrate on the MgO and ZnO-doped substrate. The diffusivity of the Na_2SO_4-NaCl molten salt on the ZnO-doped substrate was particularly outstanding. In addition, the diffusion of the molten salt on its surface indicated an enhanced trend with the increased doping concentration of MgO and ZnO in the CaO substrate. These results present that the temperature increased and ZnO and MgO doping in the CaO substrate could significantly improve the ionic mobility of the Na_2SO_4-NaCl molten salt, intensifying the deformation of Na_2SO_4-NaCl and leading to further wettability enhancement. Furthermore, the diffusion of the Na_2SO_4-NaCl salt became increasingly evident with the increase of MgO and ZnO doping concentration in the CaO substrate, promoting the wettability of molten salt.

3.3.2. Radial distribution function (RDF)

As the CaO substrates were doped with different concentrations of MgO and ZnO, the wettability of Na_2SO_4-NaCl molten salt on these substrates was enhanced. To gain a deeper insight into the impact of the structural features of the oxide substrate on the wettability of the molten Na_2SO_4-NaCl, using the RDF allows for an understanding of the internal structure of molten salt and the CaO-based substrate under high-temperature conditions. The RDF is calculated by Eq. (9) [35].

$$g(r) = \frac{1}{\rho 4\pi r^2} \frac{\sum_{\tau=1}^{\tau} \sum_{j=1}^{N} \Delta N(r-r+\delta r)}{N \times \tau} \tag{9}$$

where N represents the number of atoms, τ represents the time step, δr represents the distance, and ΔN represents the number of atoms in the range from r to δr. The peak variation of the RDF is capable of an in-depth analysis of the structural characteristics of the calcium-based oxide substrate and the ionic distribution of molten Na_2SO_4-NaCl.

Fig. 11 shows the RDF of $Ca^{1.36+}$- $Ca^{1.36+}$, $Ca^{1.36+}$- $O^{1.36-}$, and $O^{1.36-}$- $O^{1.36-}$ at different temperatures and different substrates. As the temperature increased, the thermal motion of all substrate atoms became more violent and left the original equilibrium position. Hence, the peak of the RDF becomes wider and shorter. Further comparison of the RDF of $Ca^{1.36+}$- $O^{1.36-}$, $Ca^{1.36+}$- $Ca^{1.36+}$, and $O^{1.36-}$- $O^{1.36-}$ at different substrates shows that the MgO-doped and ZnO-doped substrates have a lower RDF peak and a wider peak than the CaO substrate. The ZnO-doped substrate had the lowest and widest RDF peak. In addition, the width grew and the peak of RDF reduced when the doping concentration of MgO and ZnO in the CaO substrate increased. These were attributed to the doping of MgO and ZnO into the CaO substrate, leading to the lattice distortion of the CaO-based substrate, which caused the enhancement of the pores of the substrate. Consequently, the molten Na_2SO_4-NaCl salt would more easily wet and promote the formation of solid solutions ($Na_{3.2}Ca_{1.8}(SO_4)_3Cl_{0.8}$) [19]. The lattice parameter of this solid solution was larger than that of CaO, which further expands the pores of the substrate. The increase of pores of the substrate improved the sintering resistance of the substrate and enhanced the effective conversion rate of CaL.

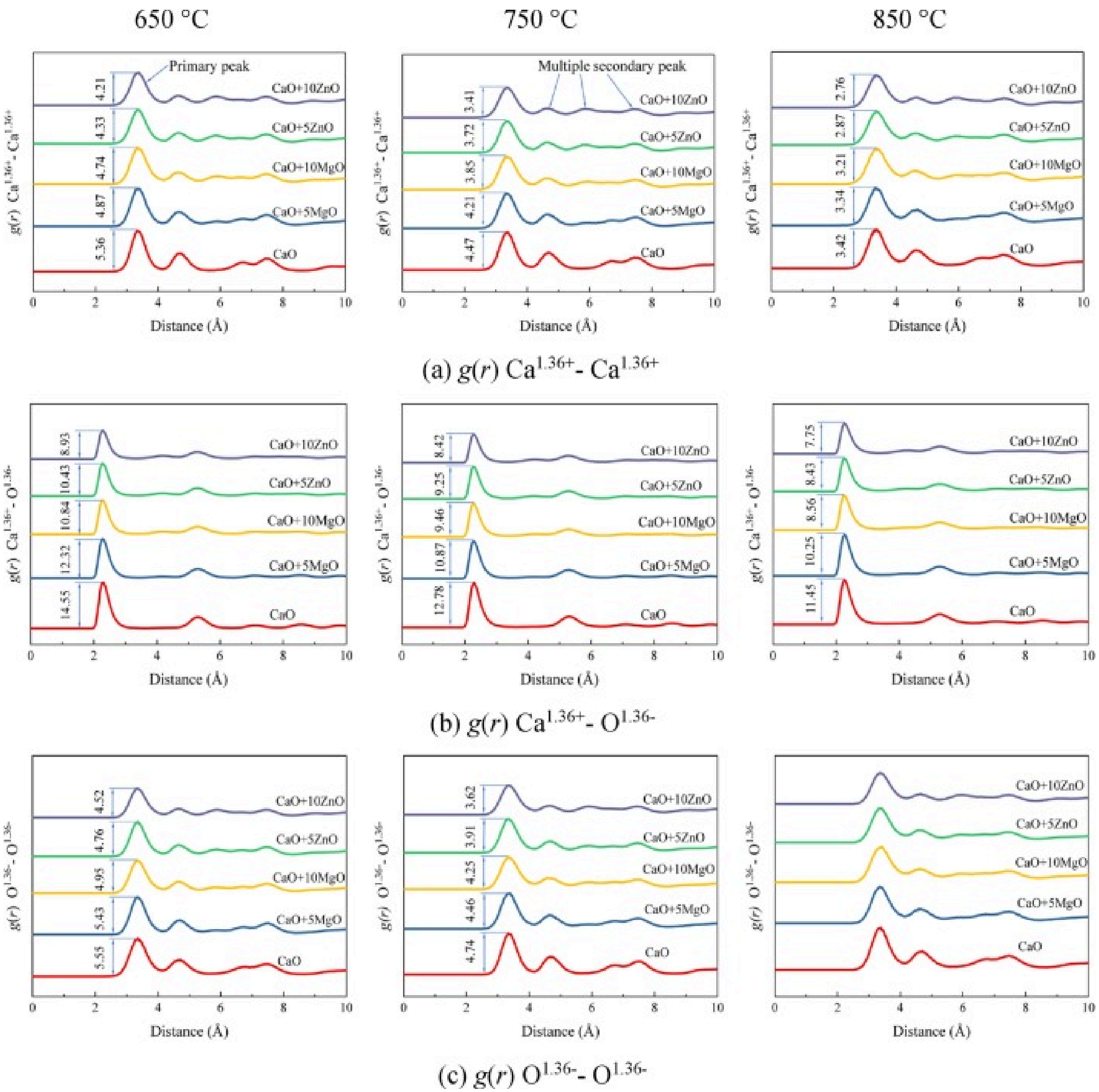

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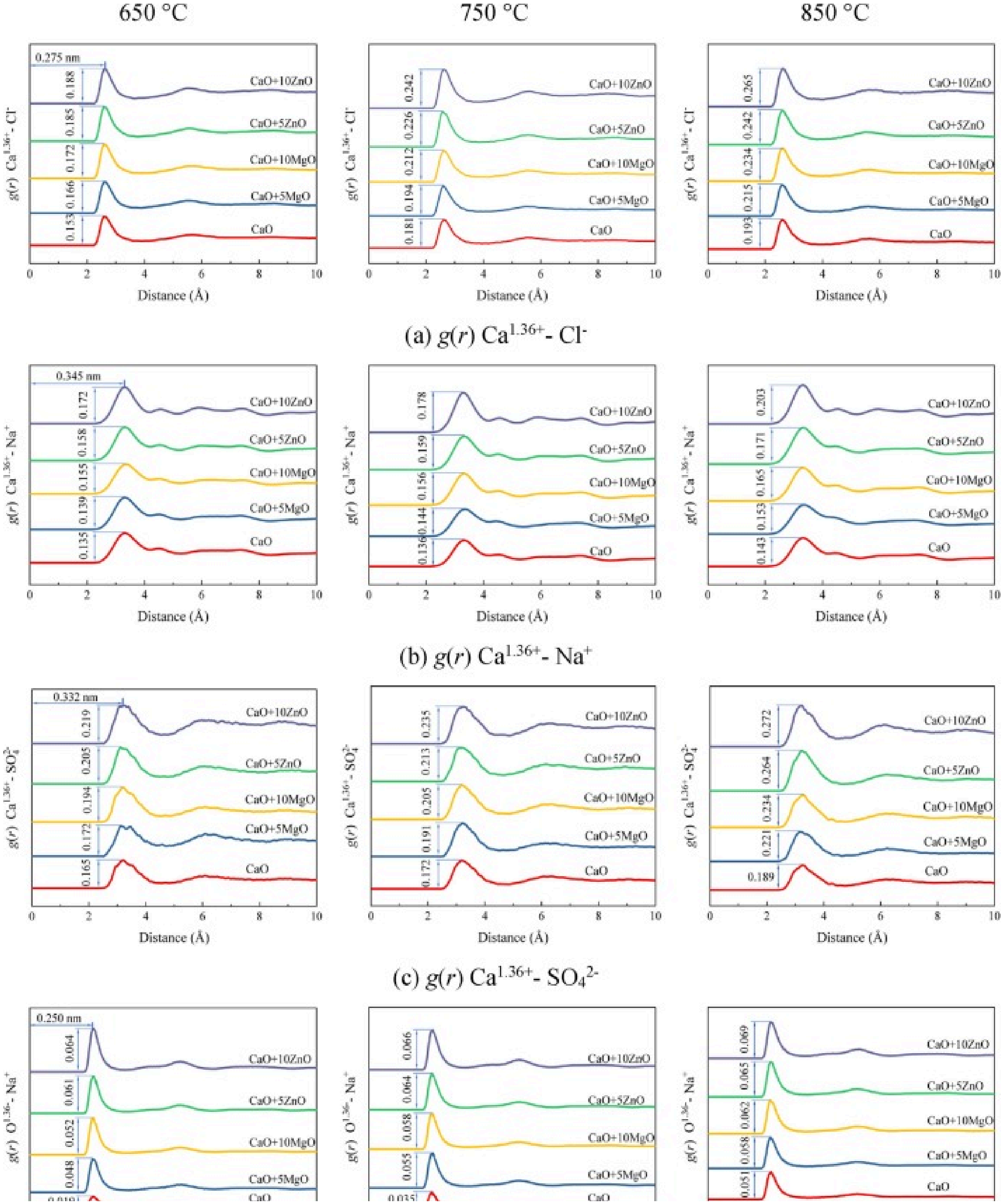
Fig. 11. RDF of (a) $Ca^{1.36+}$- $Ca^{1.36+}$, (b) $Ca^{1.36+}$- $O^{1.36-}$ and (c) $O^{1.36-}$- $O^{1.36-}$ at 650–850°C.

Fig. 12 shows the RDF of $Ca^{1.36+}$- Cl^-, $Ca^{1.36+}$- Na^+, $Ca^{1.36+}$- SO_4^{2-}, and $O^{1.36-}$- Na^+ at different temperatures and different substrates. As the temperature increased, the peak of RDF increased, showing that the frequency of sodium ions (Na^+), sulfate ions (SO_4^{2-}), and chloride ions (Cl^-) around the surface of all CaO-based substrates increased significantly. Further comparison of the RDF of $Ca^{1.36+}$- Na^+, $Ca^{1.36+}$- Cl^-, $Ca^{1.36+}$- SO_4^{2-}, and $O^{1.36-}$- Na^+ shows that the MgO-doped and ZnO-doped substrates had higher RDF peaks than the CaO substrate, indicating the ions of Na_2SO_4-NaCl salt appeared more frequently on their surface than on the surface of CaO substrate. The peaks of RDF were highest in the ZnO-doped substrate, which had the greatest amount of Na_2SO_4-NaCl ions visible on its surface. In addition, with the increased doping concentration of MgO and ZnO in the CaO substrate, the ions of Na_2SO_4-NaCl salt appeared more frequently on the surface of the substrate. These findings indicate that the temperature increase and doping of ZnO and MgO in the CaO substrate could enhance the interaction between the molten salt and the substrate and effectively improve the wettability of Na_2SO_4-NaCl. It was conducive to forming the solid solution ($Na_{3.2}Ca_{1.8}(SO_4)_3Cl_{0.8}$), which improved the reaction kinetics of CaL to advance the effective conversion rate of the CaL [19].


650 °C
750 °C
850 °C
0.275 nm
0.188
0.185
0.172
0.166
0.153
0.242
0.226
0.212
0.194
0.181
0.265
0.242
0.234
0.215
0.193
CaO+10ZnO
CaO+5ZnO
CaO+10MgO
CaO+5MgO
CaO
$g(r)$ $Ca^{1.36+}$- Cl^-
Distance (Å)
(a) $g(r)$ $Ca^{1.36+}$- Cl^-
0.345 nm
0.172
0.158
0.155
0.139
0.135
0.178
0.159
0.156
0.144
0.136
0.203
0.171
0.165
0.153
0.143
$g(r)$ $Ca^{1.36+}$- Na^+
(b) $g(r)$ $Ca^{1.36+}$- Na^+
0.332 nm
0.219
0.205
0.194
0.172
0.165
0.235
0.213
0.205
0.191
0.172
0.272
0.264
0.234
0.221
0.189
$g(r)$ $Ca^{1.36+}$- SO_4^{2-}
(c) $g(r)$ $Ca^{1.36+}$- SO_4^{2-}
0.250 nm
0.064
0.061
0.052
0.048
0.066
0.064
0.058
0.055
0.069
0.065
0.062
0.058
0.051
$g(r)$ $O^{1.36-}$- Na^+

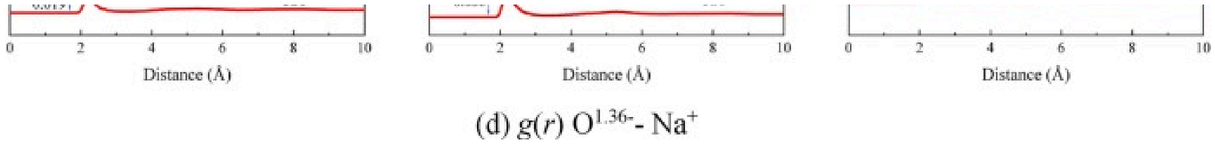


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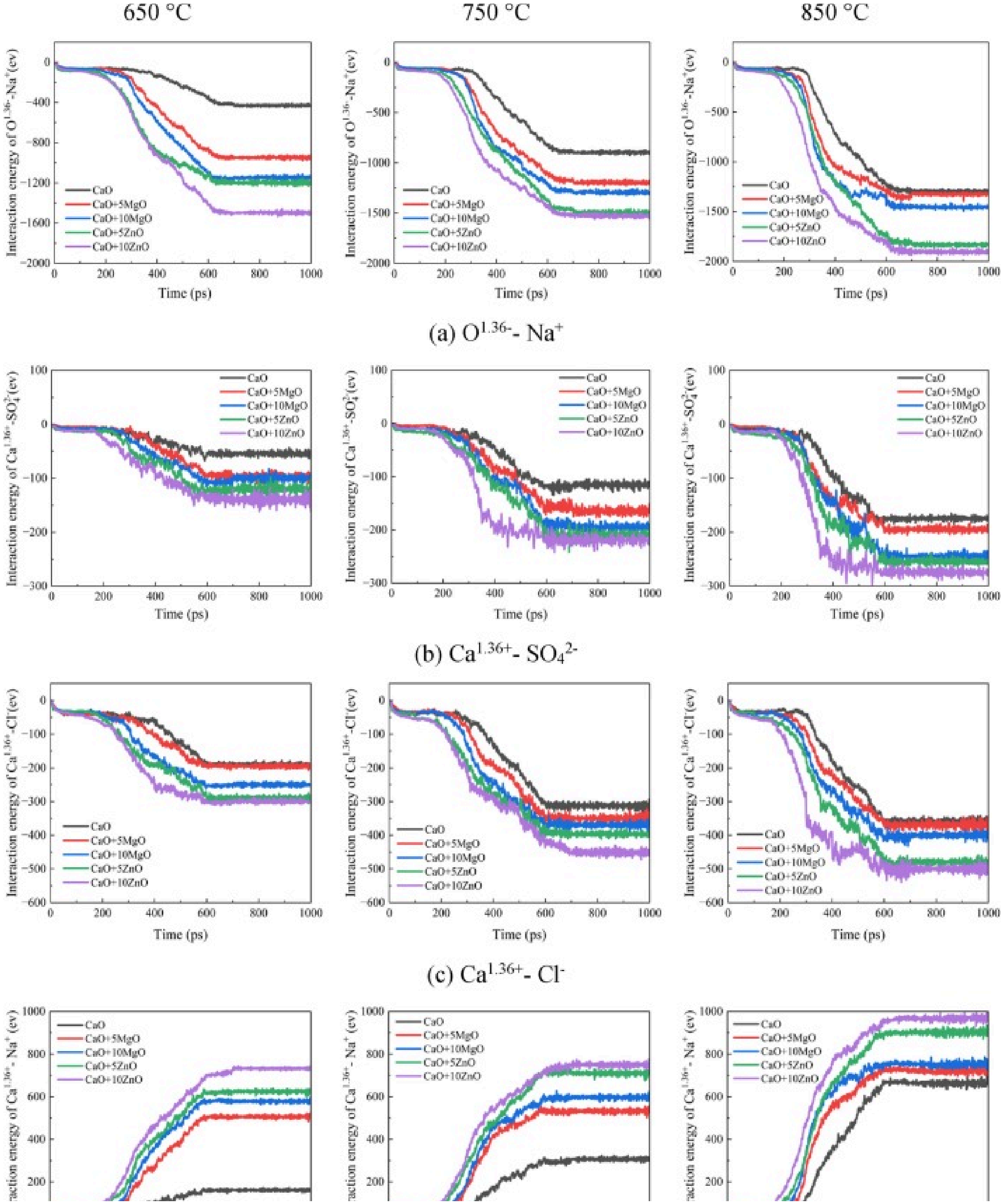
Fig. 12. RDF of (a) $Ca^{1.36+}$- Cl^{-}, (b) $Ca^{1.36+}$- Na^{+}, (c) $Ca^{1.36+}$- SO_4^{2-} and (d) $O^{1.36-}$- Na^{+} at 650–850°C.

3.3.3. Interaction energy

To demonstrate that the doping of MgO and ZnO into the CaO substrate indeed enhances the interaction between the molten salt and the substrate, Fig. 13 displays the interaction energy of $O^{1.36-}$- Na^{+}, $Ca^{1.36+}$- SO_4^{2-}, $Ca^{1.36+}$- Cl^{-}, and $Ca^{1.36+}$-Na^{+} at different temperatures and various substrates. The interaction energy between Na_2SO_4-NaCl and all substrates became stronger as the temperature increased. Further comparison of interaction energy revealed that doping MgO and ZnO in the CaO substrate increased the interaction energies and boosted with the MgO and ZnO doping concentration rates. Furthermore, the interaction energy of Na_2SO_4-NaCl and ZnO-doped substrate was highest. These conclusions once again demonstrate that doping MgO and ZnO in the CaO substrate enhanced the interaction between the Na_2SO_4-NaCl molten salt and the substrate, which was more favorable to the formation of the solid solution ($Na_{3.2}Ca_{1.8}(SO_4)_3Cl_{0.8}$).


650 °C
750 °C
850 °C
Interaction energy of O1.36- -Na+ (ev)
Time (ps)
CaO
CaO+5MgO
CaO+10MgO
CaO+5ZnO
CaO+10ZnO
(a) O1.36- - Na+
Interaction energy of Ca1.36+ -SO42- (ev)
(b) Ca1.36+ - SO42-
Interaction energy of Ca1.36+ -Cl- (ev)
(c) Ca1.36+ - Cl-

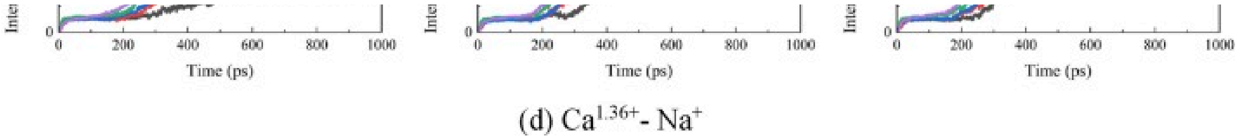


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Fig. 13. The interaction energy of (a) $O^{1.36-}$- Na^+, (b) $Ca^{1.36+}$- SO_4^{2-}, (c) $Ca^{1.36+}$- Cl^- and (d) $Ca^{1.36+}$- Na^+ at 650–850°C.

4. Conclusion

In this work, in-situ experiments were utilized to study the wettability of Na_2SO_4-NaCl binary alkali metal molten salt on CaO substrates doped with MgO and ZnO at high temperatures, and the underlying microscopic mechanism was investigated by MD simulations. The research findings show that in the temperature range of 650–850 °C, Na_2SO_4-NaCl exhibited excellent wettability on all CaO-based substrates. As the temperature increased, the interaction energy between the molten salt and the substrate was enhanced, and the contact angle between the molten Na_2SO_4-NaCl and the substrate decreased, resulting in better wettability. Doping ZnO and MgO into the CaO substrate produced lattice distortion in the substrate, which increased the pores and significantly enhanced the wettability of the molten salt on the substrate. The improvement effect increased with the increase of dopant ZnO and MgO in CaO. And the ZnO dopants demonstrated the most pronounced effect on wetting performance. More importantly, doping could enhance the interaction energy between the molten salt and the substrate, further improving the wettability of Na_2SO_4-NaCl molten salt. The enhancement of the wettability of the molten salt will facilitate the formation of solid solution, increasing the effective conversion rate of CaL. The conclusions of this study provide a scientific basis for research on improving the conversion of CaO-based materials.

CRediT authorship contribution statement

Qicheng Chen: Writing – review & editing, Funding acquisition, Conceptualization. **Qila Sa:** Software, Formal analysis, Data curation. **Nan He:** Writing – review & editing, Methodology, Investigation. **Liang Yao:** Visualization, Validation, Data curation. **Binjian Nie:** Software, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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